Predicting Crude Oil Preheat Train Fouling Using Machine Learning: A Physics-Informed Comparative Study

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Abstract

Crude-oil preheat trains are important for refinery energy efficiency but are strongly affected by heat-exchanger fouling, which reduces heat recovery and can increase operating requirements. This study investigates machine-learning prediction of crude-oil fouling rate using a physics-informed synthetic benchmark derived from an Ebert-Panchal-style threshold formulation. A dataset of 3,000 observations was generated from film temperature, velocity, tube diameter, crude-oil density and fouling propensity, with temperature-dependent viscosity used to calculate Reynolds number and wall shear stress. Four regression models—Linear Regression, Random Forest Regression, Support Vector Regression (SVR), and a Multilayer Perceptron (MLP)—were evaluated using an 80:20 train-test split and five-fold cross-validation. On the random test set, R^2 values were 0.7227 for Linear Regression, 0.9778 for Random Forest, 0.9893 for SVR, and 0.9883 for MLP. The derived hydrodynamic feature representation produced a small but consistent improvement over the primitive-variable representation. Random Forest permutation analysis identified film temperature as the most influential predictor, followed by fouling propensity, wall shear stress, and Reynolds number. A separate high-temperature holdout demonstrated that strong random-split accuracy does not guarantee reliable extrapolation: the MLP achieved R^2 = 0.7495 and SVR R^2 = 0.5519 on the unseen 566–620 K range, while Linear Regression and Random Forest produced negative R^2 values. Physics-consistency analysis showed strong agreement with the expected effects of temperature, fouling propensity and wall shear stress, while the Reynolds-number response was comparatively weak. The results demonstrate the value of combining mechanistic structure with machine learning, while emphasizing that the synthetic benchmark is not a substitute for validation using measured refinery data.

Keywords: crude-oil fouling; preheat train; machine learning; Ebert-Panchal model; support vector regression; random forest; multilayer perceptron; physics-informed prediction.

Contents

Abstract **2**

1. Introduction **5**

2. Literature Review **8**

2.1 Crude Oil Preheat Trains and Their Role in Petroleum Refining 8

2.2 Fouling Mechanisms in Crude Oil Heat Exchangers 9

2.3 Mechanistic Models for Predicting Crude Oil Fouling 11

2.4 Machine Learning Approaches to Heat Exchanger Fouling Prediction 13

2.5 Machine Learning Applications in Crude Oil Preheat Train Fouling Prediction 15

2.6 Critical Synthesis and Research Gap 17

3. Methodology **18**

3.1 Research Design 18

3.2 Physics-Informed Synthetic Dataset 19

3.3 Temperature-Dependent Viscosity 20

3.4 Reynolds Number and Wall Shear Stress 21

3.5 Physics-Informed Fouling Model 22

3.6 Addition of Synthetic Observation Noise 23

3.7 Machine-Learning Input Variables 23

3.8 Data Preprocessing 24

3.9 Machine-Learning Algorithms 24

3.10 Hyperparameter Selection 26

3.11 Model Evaluation 26

3.12 Five-Fold Cross-Validation 27

3.13 Residual Analysis 27

3.14 Feature-Importance Analysis 27

3.15 Physics-Consistency Analysis 28

3.16 Generalization to Unseen Operating Conditions 29

3.17 Derived- versus Primitive-Variable Comparison 29

3.18 Reproducibility and Computational Implementation 30

4. RESULTS **31**

4.1 Overview 31

4.2 Characteristics of the Synthetic Dataset 31

4.3 Model Performance on the Random Test Set 32

4.4 Predicted versus Actual Fouling Rate 33

4.5 Cross-Validation Results 34

4.6 Comparison of Model Performance 35

4.7 Residual Analysis 35

4.8 Random Forest Feature Importance 36

4.9 Derived versus Primitive Features 37

4.10 Physics-Consistency Analysis 38

4.11 Generalization to an Unseen Hot Regime 39

4.12 Summary of Major Findings 40

5. DISCUSSION..... 41
5.1 Overview.....41
5.2 Superiority of Nonlinear Models.....41
5.3 Why SVR Performed Best Within the Sampled Domain.....42
5.4 Importance of Film Temperature.....43
5.5 Role of Fouling Propensity.....43
5.6 Role of Wall Shear Stress.....44
5.7 The Reynolds-Number Finding.....44
5.8 Effect of Physics-Based Feature Engineering.....45
5.9 Interpolation versus Extrapolation.....46
5.10 Implications for Industrial Application.....46
5.11 Significance of the Study.....47
5.12 Limitations.....48
5.13 Future Research.....48
5.14 Chapter Summary.....49
6. CONCLUSION, RECOMMENDATIONS, LIMITATIONS AND FUTURE WORK.....50
6.1 Conclusion.....50
6.2 Recommendations.....52
6.3 Limitations of the Study.....53
6.4 Future Work.....53
6.5 Chapter Summary.....54
References..... 56

1. Introduction

Heat integration is a fundamental strategy for improving energy efficiency in petroleum refineries, where large quantities of thermal energy are continuously exchanged among process streams. One of the most important applications of this principle is the **crude oil preheat train (CPT)**, a network of shell-and-tube heat exchangers that recovers heat from hot product and pumparound streams to increase the temperature of incoming crude oil before it enters the atmospheric distillation column. By recovering energy that would otherwise be rejected, the preheat train significantly reduces furnace duty, lowers fuel consumption, decreases greenhouse gas emissions, and improves overall refinery economics. Consequently, the thermal performance of crude oil preheat trains has a direct influence on refinery profitability, energy efficiency, and operational reliability.

Despite their importance, crude oil preheat trains are persistently affected by **heat exchanger fouling**, one of the most challenging operational problems in the refining industry. Fouling refers to the unwanted accumulation of deposits on heat transfer surfaces during operation. In crude oil systems, these deposits typically arise from complex interactions among chemical reaction products, asphaltenes, inorganic salts, corrosion products, and suspended particulates. As deposits accumulate, an additional thermal resistance develops between the process fluid and the heat transfer surface, reducing the overall heat transfer coefficient and increasing the pressure drop across the exchanger. The immediate consequences include reduced heat recovery, increased furnace fuel requirements, higher pumping costs, and, ultimately, periodic shutdowns for mechanical or chemical cleaning. Beyond these direct operational penalties, fouling also accelerates equipment degradation and reduces process flexibility. Owing to its widespread occurrence and substantial economic impact, heat exchanger fouling has remained one of the most extensively investigated problems in thermal engineering for more than six decades.

The economic implications of fouling are considerable. Previous studies have estimated that heat exchanger fouling imposes substantial economic costs on the petroleum refining industry through increased energy consumption, production losses, maintenance expenses, reduced equipment availability, and unplanned shutdowns. Although routine cleaning can restore exchanger performance, cleaning schedules are frequently determined using conservative maintenance intervals or operator experience rather than accurate predictions of fouling development. Cleaning too early increases maintenance costs and production interruptions, whereas cleaning too late results in excessive energy losses and potential equipment damage. Reliable prediction of fouling behaviour is therefore essential for optimizing maintenance scheduling, improving heat recovery, and maximizing refinery profitability.

Historically, fouling prediction has relied primarily on **mechanistic and semi-empirical models** derived from the underlying physics of deposit formation and removal. The pioneering work of Kern and Seaton established the conceptual framework in which fouling is represented as the balance between competing deposition and removal processes. Building upon this foundation, Ebert and Panchal introduced the widely adopted **threshold fouling model**, specifically developed for crude oil heat exchangers. Their formulation relates fouling behaviour to the competition between temperature-dependent chemical deposition and velocity-dependent removal caused by wall shear stress. The threshold model remains one of the most influential mechanistic approaches because it captures the essential physical processes governing crude oil fouling while requiring relatively few empirical parameters.

Although mechanistic models provide valuable physical insight, they possess inherent limitations. Their predictive capability depends on empirical coefficients obtained under specific operating conditions, and these parameters often vary significantly with crude oil composition, refinery configuration, and operating practice. Modern refineries process increasingly diverse crude blends whose chemical characteristics differ substantially from those used to calibrate classical fouling models. Consequently, models that perform well under one set of operating conditions may exhibit reduced accuracy when applied to different crude sources or changing process conditions. Furthermore, the nonlinear interactions among operating variables such as temperature, Reynolds number, wall shear stress, residence time, and crude composition are difficult to represent using simplified analytical expressions.

Advances in industrial data acquisition and computational methods have created new opportunities for machine learning (ML) as a complementary approach to fouling prediction. Unlike mechanistic models, which require explicit mathematical descriptions of physical processes, machine learning algorithms identify relationships directly from data. By learning complex nonlinear interactions among multiple operating variables, ML models can often achieve high predictive accuracy without requiring detailed knowledge of the underlying fouling mechanisms. Over the past two decades, artificial neural networks, support vector machines, decision-tree ensembles, Gaussian process regression, and deep learning architectures have been increasingly applied to fouling prediction across a variety of industrial heat exchanger systems. Recent reviews and comparative studies report promising predictive performance for several ML approaches, while also emphasizing the importance of data quality, model validation, and the operating range represented in the training data (Soomro et al., 2026).

Within crude oil preheat train applications, machine learning has demonstrated particularly encouraging results. Studies employing artificial neural networks have reported substantially lower prediction errors than classical threshold fouling models when estimating crude oil fouling rates. More recent investigations have expanded these approaches by incorporating ensemble learning algorithms, recurrent neural networks, and hybrid prediction frameworks

capable of exploiting increasingly large volumes of process data generated by modern refinery instrumentation. Despite these advances, a significant practical limitation remains: most published machine learning studies rely on proprietary industrial datasets that are unavailable to independent researchers. This lack of publicly accessible data limits reproducibility, makes objective comparison among algorithms difficult, and creates barriers for researchers seeking to evaluate alternative prediction methods.

The present study addresses the limitations associated with data availability and controlled model comparison by developing a physics-informed benchmark dataset using an Ebert-Panchal-style threshold fouling formulation with illustrative parameters. Rather than replacing mechanistic understanding, the synthetic dataset preserves selected physical relationships between deposition and removal while providing a transparent and reproducible basis for evaluating machine-learning algorithms. Four regression algorithms—Linear Regression, Random Forest Regression, Support Vector Regression (SVR), and a Multilayer Perceptron (MLP)—are evaluated within the same dataset and validation framework. The comparison considers both linear and nonlinear approaches without treating any one model as the primary benchmark.

The objectives of this study are therefore to: (i) review mechanistic and machine-learning approaches for crude oil preheat train fouling prediction; (ii) generate a reproducible physics-informed synthetic dataset using an Ebert-Panchal-style threshold formulation; (iii) evaluate Linear Regression, Random Forest Regression, Support Vector Regression and a Multilayer Perceptron using an 80:20 train-test split and five-fold cross-validation; (iv) compare models using primitive operating variables with models using derived hydrodynamic features, including Reynolds number and wall shear stress; and (v) assess feature importance, consistency with expected physical trends and generalization to an unseen high-temperature regime. By integrating established heat-transfer and hydrodynamic concepts with machine learning, this work presents a controlled benchmarking framework while recognizing the limitations of synthetic data for representing refinery fouling.

2. Literature Review

2.1 Crude Oil Preheat Trains and Their Role in Petroleum Refining

Heat integration represents one of the most effective strategies for reducing energy consumption in petroleum refining. Among the numerous heat recovery systems employed within a refinery, the crude oil preheat train (CPT) occupies a central position because it recovers thermal energy from hot process streams before crude oil enters the atmospheric distillation column. Instead of relying solely on fired heaters to achieve the required feed temperature, the crude preheat train transfers waste heat from product streams, pumparounds, and side draws to the incoming crude through a series of shell-and-tube heat exchangers arranged in series and parallel. This network substantially reduces furnace duty, lowers fuel consumption, decreases carbon emissions, and improves the overall thermal efficiency of the refinery.

The economic significance of crude preheat trains extends beyond energy conservation. Because furnace firing accounts for a considerable proportion of refinery operating costs, even modest reductions in heat recovery efficiency can translate into substantial increases in energy expenditure. Consequently, maintaining high heat transfer performance throughout the preheat train is essential for achieving optimal refinery profitability. Industrial studies consistently identify the crude preheat train as one of the largest opportunities for improving refinery energy efficiency because its performance directly influences downstream separation efficiency, fuel consumption, and process throughput.

Despite careful design and optimization, the thermal performance of crude preheat trains gradually deteriorates during operation owing to the accumulation of deposits on heat transfer surfaces. This degradation is unavoidable because crude oil is an exceptionally complex mixture containing hydrocarbons, resins, asphaltenes, inorganic salts, corrosion products, suspended solids, and numerous trace contaminants. Under refinery operating conditions, these components interact through chemical reactions, precipitation, and physical deposition, progressively forming insulating layers on exchanger surfaces. As the fouling layer thickens, thermal resistance increases while the effective flow area decreases, leading simultaneously to reduced heat transfer coefficients and increased pressure losses.

The consequences of fouling extend throughout refinery operations. Reduced heat recovery requires additional furnace firing to achieve the desired distillation feed temperature, increasing fuel consumption and greenhouse gas emissions. Increased pressure drop raises pumping power requirements and may alter flow distribution throughout the exchanger network. If left unmanaged, excessive fouling eventually necessitates costly shutdowns for exchanger cleaning, resulting in production losses that often exceed the direct maintenance

costs. Because refineries typically operate continuously for extended periods, determining the optimal timing for cleaning interventions represents a significant operational and economic challenge.

The cumulative impact of these effects explains why crude oil fouling has remained a major research topic for several decades. Previous studies have estimated substantial annual economic costs associated with heat exchanger fouling in the petroleum refining industry, reflecting increased energy consumption, maintenance expenditure, reduced equipment availability, and production losses. Consequently, reliable prediction of fouling development has become an important objective for refinery operators seeking to transition from scheduled maintenance toward predictive and condition-based maintenance strategies.

2.2 Fouling Mechanisms in Crude Oil Heat Exchangers

Although fouling is often discussed as a single phenomenon, it actually encompasses several distinct deposition mechanisms that may occur independently or simultaneously depending on fluid composition and operating conditions. Understanding these mechanisms is essential because predictive models must ultimately represent the physical processes responsible for deposit formation.

One of the most significant mechanisms in crude oil systems is **chemical reaction fouling**, frequently referred to as coking. At elevated surface temperatures, unstable hydrocarbon molecules undergo thermal cracking, oxidation, polymerization, and condensation reactions that produce high-molecular-weight compounds with limited solubility. These compounds gradually adhere to heat transfer surfaces, forming carbonaceous deposits that continue to grow as additional reaction products accumulate. Chemical reaction fouling is particularly severe in refinery preheat trains because wall temperatures frequently approach the range in which thermal degradation reactions become increasingly significant.

Another important mechanism is **precipitation fouling**, which occurs when dissolved species become insoluble as operating conditions change. Variations in temperature, pressure, or chemical composition may reduce the solubility of inorganic salts or heavy organic fractions such as waxes and asphaltenes, causing them to precipitate onto exchanger surfaces. In crude oil processing, asphaltene precipitation has received particular attention because crude blending, temperature variation, and compositional instability can all promote deposit formation. Unlike purely chemical fouling, precipitation is strongly influenced by thermodynamic equilibrium and fluid composition rather than reaction kinetics alone.

Particulate fouling represents a third major mechanism encountered in refinery operation. Suspended solids originating from corrosion products, catalyst fines, sand, coke particles, or upstream process equipment may settle on heat transfer surfaces when fluid velocities become

insufficient to maintain complete suspension. The probability of particulate deposition therefore depends strongly on hydrodynamic conditions, particularly Reynolds number, turbulence intensity, and wall shear stress. High-velocity turbulent flow tends to remove loosely attached particles, whereas low flow velocities promote deposition by reducing the forces available to detach accumulated material.

Corrosion processes may also contribute directly to fouling through the formation of corrosion products that remain attached to exchanger surfaces or become incorporated into existing deposits. In many practical situations, corrosion and fouling occur simultaneously, producing complex deposits whose composition evolves continuously throughout the operating cycle. Biological fouling, while important in cooling-water systems, generally plays a much less significant role within crude oil preheat trains because operating temperatures are typically too high to support microbial activity.

In practice, fouling rarely results from a single isolated mechanism. Instead, deposit formation reflects the interaction of chemical reactions, precipitation phenomena, particulate transport, hydrodynamic conditions, and surface properties. These interactions explain why fouling behaviour often exhibits pronounced nonlinear characteristics. For example, increasing temperature accelerates chemical reaction rates and promotes coke formation, whereas increasing flow velocity simultaneously enhances heat transfer while increasing wall shear stress capable of removing deposits. The resulting balance between deposition and removal determines the net fouling rate observed during operation.

Experimental investigations consistently demonstrate that **temperature and flow velocity** exert the greatest influence on crude oil fouling behaviour. Elevated temperatures generally accelerate deposit formation by increasing reaction rates, whereas higher Reynolds numbers reduce fouling through increased turbulence and greater wall shear stress. These competing effects underpin many mechanistic fouling models developed over the past several decades and likewise explain why these variables frequently emerge as the most important predictors in machine learning studies. Their repeated identification across independent investigations suggests that successful predictive models must adequately represent both the thermally activated processes responsible for deposit formation and the hydrodynamic mechanisms governing deposit removal.

The complexity arising from these coupled physical and chemical processes also highlights the limitations of purely empirical correlations. Models based exclusively on curve fitting often fail when operating conditions move beyond the range of experimental observations because they lack direct representation of the governing physics. Conversely, purely mechanistic formulations may oversimplify the interactions among multiple fouling mechanisms by assuming idealized relationships between operating variables. This balance between physical

interpretability and predictive flexibility has motivated the continued evolution of fouling models from classical mechanistic formulations toward modern data-driven approaches capable of learning complex nonlinear relationships directly from operational data.

2.3 Mechanistic Models for Predicting Crude Oil Fouling

The earliest attempts to predict heat exchanger fouling were based on mechanistic descriptions of the physical processes governing deposit formation and removal. Unlike purely empirical correlations, mechanistic models seek to represent the underlying transport phenomena responsible for fouling through equations derived from heat transfer, fluid mechanics, and chemical reaction engineering. Although numerous mechanistic formulations have been proposed over the past six decades, two models have had particularly enduring influence in petroleum refining: the **Kern-Seaton model** and the **Ebert-Panchal threshold fouling model**.

The work of **Kern and Seaton (1959)** established the conceptual foundation upon which most subsequent fouling models have been developed. They proposed that net fouling growth results from the competition between two simultaneous processes: deposition of foulant material onto the heat transfer surface and removal of previously deposited material by hydrodynamic forces. Under this framework, the observed fouling rate is expressed as the difference between deposition and removal rates. Initially, deposition dominates and the fouling layer grows rapidly; however, as deposits accumulate, removal becomes increasingly significant until the two processes eventually approach equilibrium. This balance explains the asymptotic fouling curves frequently observed in industrial heat exchangers, where fouling resistance increases rapidly during early operation before approaching a limiting value.

The principal strength of the Kern-Seaton formulation lies in its physical interpretability. Unlike empirical curve-fitting methods, it provides a mechanistic explanation for why fouling eventually stabilizes rather than increasing indefinitely. Moreover, the deposition-removal framework can be adapted to different industrial systems by modifying the mathematical expressions describing each process. Consequently, the Kern-Seaton concept continues to underpin many modern fouling models despite its relatively simple mathematical formulation.

Nevertheless, the original model was intentionally generic and does not explicitly account for the complex chemistry associated with crude oil fouling. Petroleum-derived deposits arise from numerous interacting mechanisms, including thermal cracking, oxidation, polymerisation, precipitation of heavy hydrocarbons, and particulate deposition. Representing these phenomena through a single deposition term inevitably limits predictive accuracy when applied to industrial crude oil systems exhibiting wide variations in composition and operating conditions.

To overcome these limitations, **Ebert and Panchal (1995)** proposed the threshold fouling model specifically for crude oil heat exchangers. Developed using Exxon crude-oil slip-stream coking data, the model introduced an important conceptual advance by recognising that fouling does not necessarily occur under all operating conditions. Instead, deposition and removal compete continuously, producing a threshold below which net fouling becomes negligible.

Within the Ebert-Panchal framework, deposition is represented by an Arrhenius-type expression that increases exponentially with film temperature, reflecting the temperature dependence of chemical reaction rates responsible for coke precursor formation. Removal, in contrast, is governed primarily by wall shear stress generated by turbulent flow. As fluid velocity increases, greater shear forces act upon deposited material, reducing the net fouling rate. The resulting model therefore captures the two dominant physical influences repeatedly observed in refinery operation: increasing temperature promotes fouling, whereas increasing Reynolds number suppresses it.

One reason for the enduring popularity of the threshold model is its consistency with industrial observations. Refineries have long recognised that relatively small increases in crude velocity can significantly reduce deposit accumulation, while modest increases in tube wall temperature often accelerate fouling disproportionately. The threshold formulation reproduces both behaviours using physically meaningful variables rather than purely statistical relationships, making it valuable not only for performance prediction but also for exchanger design, operating optimisation, and sensitivity analysis.

Despite these advantages, the Ebert-Panchal model remains subject to several important limitations. The empirical constants contained within the deposition term must be determined experimentally and often vary according to crude source, blending strategy, operating history, and refinery configuration. Consequently, parameter values calibrated for one refinery may not transfer reliably to another processing chemically distinct crude oils. Furthermore, the model simplifies numerous interacting chemical pathways into a relatively compact mathematical expression, necessarily neglecting compositional changes that occur during prolonged refinery operation.

Another limitation arises from the deterministic nature of mechanistic modelling itself. Classical models assume that the governing physical relationships remain constant throughout operation, whereas industrial refineries frequently experience fluctuating feed compositions, varying production schedules, equipment ageing, and transient operating conditions. These sources of variability introduce nonlinear interactions that are difficult to capture using analytical equations alone.

Nevertheless, mechanistic models remain indispensable because they provide physical insight unavailable from purely statistical approaches. Their parameters possess direct engineering

meaning, allowing operators to understand how changes in temperature, velocity, or wall shear influence fouling development. For this reason, modern research increasingly views mechanistic and data-driven approaches as complementary rather than competing methodologies. Mechanistic models provide the physical framework, whereas machine learning offers the flexibility required to capture complex nonlinear behaviour that exceeds the simplifying assumptions of analytical formulations.

2.4 Machine Learning Approaches to Heat Exchanger Fouling Prediction

The rapid expansion of industrial sensing technologies, distributed control systems, and process historians has transformed modern chemical plants into data-rich environments capable of recording thousands of operating variables continuously. This unprecedented availability of process data has accelerated the application of **machine learning (ML)** to engineering problems that are difficult to model using first-principles equations alone. Heat exchanger fouling represents one such problem, owing to its dependence on numerous interacting variables whose relationships are highly nonlinear and often poorly understood.

Unlike mechanistic models, machine learning algorithms do not require explicit mathematical descriptions of deposition mechanisms. Instead, they identify statistical relationships directly from historical observations, enabling prediction even when the underlying physical processes remain only partially understood. This characteristic makes machine learning particularly attractive for crude oil fouling, where deposit formation depends simultaneously on thermal conditions, fluid properties, hydrodynamics, crude composition, exchanger geometry, and operating history.

Among the earliest machine learning techniques applied to fouling prediction were artificial neural networks (ANNs). Inspired by the structure of biological neural systems, ANNs consist of interconnected computational nodes capable of approximating highly nonlinear functions through iterative training. Because neural networks can learn complex interactions among multiple variables without predefined equations, they became widely used for predicting heat exchanger performance and fouling. Studies have reported that ANN models can provide accurate fouling predictions when sufficient and representative training data are available (Davoudi & Vaferi, 2018). However, neural networks are frequently criticised for their limited interpretability, as their internal decision-making processes are difficult to relate directly to physical engineering principles.

More recently, **ensemble learning algorithms** have gained increasing attention within heat transfer research. Random Forest Regression, one of the most widely adopted ensemble techniques, constructs a large collection of decision trees using randomly selected subsets of

both observations and predictor variables. Final predictions are obtained by averaging the outputs of individual trees, substantially reducing prediction variance while improving generalisation performance. Unlike single decision trees, Random Forest models are relatively resistant to overfitting and provide quantitative measures of feature importance, enabling researchers to identify the operating variables that most strongly influence fouling behaviour. This balance between predictive accuracy and interpretability has contributed significantly to their growing popularity in chemical engineering applications.

Gradient-boosting methods, including **XGBoost** and **LightGBM**, represent another major development in data-driven fouling prediction. Rather than averaging independently constructed trees, boosting algorithms sequentially train each new tree to correct errors made by previous models, often producing superior predictive performance on structured engineering datasets. Several recent studies have reported that boosted tree algorithms achieve lower prediction errors than conventional regression models while maintaining reasonable computational efficiency. Their principal disadvantage lies in the increased complexity associated with parameter tuning, which may require extensive optimisation before satisfactory performance is achieved.

Support Vector Regression (SVR) has also been successfully applied to nonlinear engineering prediction problems. By constructing an optimal regression function within a high-dimensional feature space, SVR can model complex relationships while maintaining good generalisation performance for relatively small datasets. However, prediction accuracy depends strongly on the choice of kernel function and associated hyperparameters, making model development more computationally demanding than conventional linear regression.

Deep learning techniques, particularly Long Short-Term Memory (LSTM) networks, have emerged as promising tools for analysing time-dependent refinery and heat-exchanger data. Unlike conventional feedforward neural networks, LSTM architectures explicitly retain information from previous observations, making them well suited to modelling gradual fouling progression recorded by continuously operating process sensors. Recent work has demonstrated deep-learning frameworks for fouling prediction and has shown that such models can learn nonlinear relationships between operating variables and fouling resistance (Sundar et al., 2020). However, their substantial data and computational requirements can limit their applicability when only limited experimental measurements are available.

Despite the diversity of machine learning algorithms investigated, several recurring observations emerge across the literature. First, nonlinear models can outperform simple linear regression when the underlying fouling relationships are strongly nonlinear. Second, temperature and flow-related variables frequently emerge as important predictors, reflecting the deposition-removal balance embedded within classical mechanistic models. Finally,

prediction accuracy depends strongly on data quality and representativeness as well as algorithm selection. Sophisticated machine learning models cannot compensate for noisy, biased, or insufficiently representative training data, highlighting the continued importance of high-quality datasets in engineering prediction.

Rather than replacing mechanistic understanding, contemporary research increasingly advocates integrating physics-based knowledge with machine learning. Such physics-informed machine learning combines relationships from mechanistic models with the predictive flexibility of data-driven methods and can support controlled evaluation where suitable operational data are unavailable. This provides the conceptual basis for the present study: an Ebert-Panchal-style threshold formulation with illustrative parameters is used to generate a transparent synthetic benchmark for evaluating four regression algorithms under reproducible conditions. The resulting benchmark represents the selected formulation and assumptions; it is not a calibrated model of a particular refinery or crude oil.

2.5 Machine Learning Applications in Crude Oil Preheat Train Fouling Prediction

Although machine learning has been successfully applied to a wide range of industrial heat transfer problems, its application to crude oil preheat train fouling presents unique challenges owing to the complexity of petroleum chemistry and the limited availability of publicly accessible operational data. Unlike cooling-water systems or laboratory-scale heat exchangers, refinery preheat trains operate under continuously changing feed compositions, flow conditions, and production schedules, resulting in highly nonlinear fouling behaviour that is difficult to describe using conventional mechanistic models alone. These characteristics have motivated increasing interest in data-driven prediction methods capable of extracting predictive relationships directly from historical operating data.

One of the earliest and most influential applications of machine learning to crude oil fouling prediction was reported by Aminian and Shahhosseini (2008), who developed a multilayer artificial neural network to predict fouling rates using variables including crude velocity, tube surface temperature, and tube diameter. Their study demonstrated a substantial improvement over three established threshold fouling models, with the neural network achieving a mean relative error of 26.23% compared with 47.9% for the best-performing mechanistic model. More importantly, sensitivity analysis identified crude velocity and tube diameter as more influential than surface temperature within the investigated operating range, illustrating the ability of machine learning not only to improve prediction accuracy but also to provide insight into the relative importance of operating variables. (Aminian & Shahhosseini, 2008).

Building upon this work, Aminian and Shahhosseini (2009) extended neural-network modelling to the prediction of the fouling threshold itself rather than fouling rate. By relating threshold behaviour to Reynolds number, Prandtl number, and surface temperature, they demonstrated

that neural networks could successfully approximate relationships traditionally represented through empirical mechanistic equations. Their findings reinforced the view that machine learning should not be regarded merely as an alternative to mechanistic modelling, but as a complementary tool capable of approximating complex nonlinear relationships that are difficult to represent analytically. (Aminian & Shahhosseini, 2009).

Subsequent developments in machine learning have expanded beyond conventional artificial neural networks. Improvements in computational power and data availability have enabled the application of ensemble learning methods such as Random Forest, Gradient Boosting, and Extreme Gradient Boosting (XGBoost), together with advanced deep-learning architectures including Long Short-Term Memory (LSTM) networks. These methods provide different forms of nonlinear modelling flexibility, although their performance depends on the characteristics and representativeness of the available data. In refinery applications where operating conditions evolve continuously over time, recurrent neural networks may be useful because they explicitly capture temporal dependencies that conventional feedforward networks do not represent.

Recent studies and reviews further demonstrate growing interest in hybrid prediction frameworks that combine multiple algorithms or integrate machine learning with physical process knowledge (Soomro et al., 2026). Rather than treating mechanistic and data-driven approaches as competing methodologies, these approaches can exploit complementary strengths. Mechanistic models contribute physically interpretable relationships based on heat transfer, fluid mechanics, and reaction engineering, whereas machine learning can represent nonlinear interactions that may be difficult to capture with simplified analytical formulations. This convergence provides the conceptual basis for the present study: an Ebert-Panchal-style threshold formulation with illustrative parameters is used to generate a transparent synthetic benchmark for evaluating four regression algorithms under reproducible conditions. The resulting benchmark represents the selected formulation and assumptions; it is not a calibrated model of a particular refinery or crude oil.

Despite the diversity of machine learning approaches investigated, several recurring observations emerge across published studies. First, nonlinear algorithms can outperform simple linear models when the underlying fouling relationships are strongly nonlinear. Second, flow velocity or Reynolds number and temperature frequently appear among the influential predictor variables. This is consistent with the physical assumptions underlying the Ebert-Panchal threshold model, where deposition is promoted by increasing temperature while removal is enhanced by wall shear associated with flow. These observations support the use of physically meaningful variables when developing data-driven fouling models.

Nevertheless, important limitations remain. Many published machine learning studies depend upon proprietary refinery measurements or laboratory datasets that are not readily accessible

to independent researchers. Consequently, reported prediction accuracies cannot always be independently verified, benchmarked, or reproduced. Differences in crude composition, exchanger geometry, operating strategy, and data preprocessing further complicate comparisons among published models, making it difficult to determine whether improvements in predictive performance arise from algorithm choice or from differences in the underlying datasets. This lack of transparency remains an important barrier to broader reproducibility in industrial fouling research.

2.6 Critical Synthesis and Research Gap

The literature reviewed demonstrates substantial progress in understanding and predicting crude oil fouling through mechanistic modelling and machine learning. Classical mechanistic approaches, particularly the Kern-Seaton and Ebert-Panchal formulations, remain valuable because they are grounded in heat transfer, reaction kinetics, and fluid mechanics. However, simplified relationships and empirically calibrated parameters can limit their application when refinery conditions differ from those used during model development. Machine-learning studies have explored linear and nonlinear methods, including neural networks, Random Forest, Gradient Boosting and Support Vector Regression, but findings across studies are difficult to compare when they rely on different or proprietary datasets and validation procedures.

First, reproducibility remains a challenge because many studies rely on proprietary industrial or laboratory datasets that cannot be independently accessed, validated or reused. This restricts controlled comparisons among algorithms and makes it difficult to distinguish algorithmic effects from differences in data and preprocessing.

Second, comparing models under the same transparent conditions is necessary to understand how linear and nonlinear approaches perform relative to one another. A useful benchmark should evaluate several algorithms using consistent data splits and complementary validation, rather than infer general performance from results obtained on separate datasets.

Third, predictive accuracy alone does not establish whether a model has captured physically plausible behaviour or will generalize beyond the conditions represented during training. The representation of operating conditions also matters: derived hydrodynamic quantities such as Reynolds number and wall shear stress may provide information that is not expressed as directly by primitive measurements alone. These considerations motivate assessment of feature representations, expected physical trends and performance under an operating regime withheld from model development.

The present study addresses these methodological gaps using a reproducible synthetic benchmark generated with an Ebert-Panchal-style threshold fouling formulation and illustrative, literature-informed parameters. Linear Regression, Random Forest Regression,

Support Vector Regression and a Multilayer Perceptron are evaluated using an 80:20 random train-test split and five-fold cross-validation. The analysis compares primitive operating variables with a feature set that includes derived Reynolds number and wall shear stress, examines feature importance and expected physical trends, and tests generalization to a held-out high-temperature range. This controlled design supports comparison of predictive behaviour within the assumptions of the synthetic formulation; it does not establish performance on plant measurements.

By integrating mechanistic concepts with data-driven prediction, this study contributes a reproducible framework for evaluating machine-learning models for crude oil preheat train fouling. Its contribution lies in considering predictive accuracy alongside feature representation, physical consistency and performance under an unseen operating regime, thereby making the limits of random-split evaluation more explicit.

3. Methodology

3.1 Research Design

This study adopted a computational, physics-informed machine-learning approach to investigate the prediction of crude-oil fouling rate in refinery preheat-train systems. The study was designed to determine whether machine-learning models could reproduce a nonlinear

fouling relationship derived from established fouling concepts and to examine how well the models generalize when exposed to operating conditions outside their training range.

Because an appropriate experimental or industrial fouling dataset was not available within the scope of the study, a synthetic dataset was generated using a physically motivated fouling-rate formulation. The synthetic dataset was not intended to represent direct refinery measurements. Instead, it provided a controlled environment in which different machine-learning algorithms could be compared while retaining physically meaningful relationships between operating variables.

Four regression algorithms were evaluated: Linear Regression, Random Forest Regression, Multilayer Perceptron (MLP) and Support Vector Regression (SVR). Model performance was assessed using an independent test set and five-fold cross-validation. Additional analyses were performed to examine feature importance, residual behaviour, physical consistency and generalization to an unseen high-temperature operating regime.

3.2 Physics-Informed Synthetic Dataset

A total of 3,000 observations were generated. Rather than independently sampling all model inputs, the revised data-generation procedure began with primitive physical variables and subsequently calculated the hydrodynamic quantities used by the machine-learning models.

The primitive variables were:

- film temperature, T_f
- tube-side velocity, v
- tube internal diameter, D
- crude-oil density, ρ
- fouling propensity index, F_p

The ranges used to generate these variables were:

Variable	Range
Film temperature	400–620 K
Velocity	0.6–2.5 m/s
Tube diameter	0.016–0.025

Variable	Range
	m
Density	800–900 kg/m³
Fouling propensity index	0.5–1.5

The fouling propensity index was introduced as a dimensionless representation of differences in the tendency of different crude-oil compositions to form deposits. A value of 1.0 represented the reference crude, while values below or above 1.0 represented comparatively lower or higher fouling propensity, respectively.

A fixed random seed of 42 was used to make the synthetic dataset and the subsequent model experiments reproducible.

3.3 Temperature-Dependent Viscosity

Because crude-oil viscosity changes substantially with temperature and directly affects flow behaviour, viscosity was calculated rather than independently sampled.

An Andrade/Arrhenius-type relationship was used:

$$\mu = \mu_0 \exp\left(\frac{E_v}{R T_f}\right)$$

where:

- $\mu$is dynamic viscosity, Pa·s
- $\mu_0$is the pre-exponential viscosity constant
- $E_v$is the viscosity activation parameter, J/mol
- $R$is the universal gas constant, 8.314J/(mol·K)
- $T_f$is film temperature, K

The parameters used were:

$$\mu_0 = 6.076 \times 10^{-6}$$

and

$$E_v = 21589\, J/mol.$$

This formulation was selected to produce a substantial reduction in viscosity over the investigated temperature range, providing a physically motivated link between temperature and hydrodynamic

3.4 Reynolds Number and Wall Shear Stress

Reynolds number was calculated from the primitive flow variables according to:

$$\Re = \frac{\rho v D}{\mu}$$

where $\Re$ is the Reynolds number.

This approach was used instead of independently sampling Reynolds number because Reynolds number is physically determined by fluid properties and flow geometry.

The resulting synthetic dataset produced Reynolds numbers ranging approximately from 2,320 to 117,815, with a mean of approximately 30,326. Because the generated range includes values near the laminar-to-transitional region, the use of the Blasius correlation across the full synthetic range represents a simplifying modelling assumption and is treated as a limitation of the synthetic benchmark.

For the synthetic benchmark, the Darcy friction factor was estimated using the Blasius correlation for smooth-tube flow:

$$f = 0.316\, Re^{-0.25}$$

Wall shear stress was then calculated using:

$$\tau_w = \frac{f \rho v^2}{8}$$

where τ_w is wall shear stress in Pa.

The resulting wall shear stresses ranged approximately from **0.98 to 20.79 Pa**, with a mean of approximately **6.98 Pa**.

This procedure created a physical relationship between Reynolds number and wall shear stress rather than treating them as statistically independent variables.

3.5 Physics-Informed Fouling Model

The synthetic fouling rate was generated using an Ebert-Panchal-inspired formulation based on the competing effects of deposition and removal:

$$\frac{dR_f}{dt} = \alpha F_p R e^{\beta} \exp\left(\frac{-E_a}{RT_f}\right) - \gamma \tau_w$$

where:

- dR_f/dt is the fouling-rate response
- α is a deposition scaling parameter
- F_p is the fouling propensity index
- $\Re$ is Reynolds number
- β is the Reynolds-number exponent
- E_a is the activation energy associated with the deposition process
- R is the universal gas constant
- T_f is film temperature
- γ is the removal coefficient
- τ_w is wall shear stress.

The parameter values used were:

$$\alpha = 3.0 \times 10^7 \quad \beta = -0.75 \quad E_a = 60\,000\ J/mol$$

and

$$\gamma = 4.0 \times 10^{-4}.$$

The negative value of β represents the reduction in the deposition contribution with increasing Reynolds number within the adopted formulation, while the wall-shear term represents deposit removal by hydrodynamic forces.

A non-negative constraint was applied to the theoretical fouling rate:

$R_f = max\left(\frac{d\,R_f}{dt}, 0\right)$ so that the generated net fouling rate could not become physically negative.

The parameter magnitudes were **literature-informed illustrative values** and were not intended to reproduce the fitted coefficients of any particular experimental study. Consequently, the formulation should be regarded as an **Ebert-Panchal-inspired physics-informed model**, rather than an exact reproduction of the original published fouling correlation.

3.6 Addition of Synthetic Observation Noise

To prevent the machine-learning models from simply learning an entirely deterministic mathematical function, random observational noise was added to the calculated fouling rates.

The noise standard deviation was defined as:

$\sigma_{\text{noise}} = 0.08 \vee R_f \vee + 2 \times 10^{-6}$ and the observed synthetic fouling rate was calculated as:

$$R_{f,\text{obs}} = max\left(R_f + \epsilon, 0\right) \text{ where}$$

$$\epsilon \sim N\left(0, \sigma^2_{\text{noise}}\right).$$

The final synthetic target had a range of approximately **0 to 0.184 m²K/kWh**, with a mean of approximately **0.0173 m²K/kWh** and a standard deviation of approximately **0.0236 m²K/kWh**.

Approximately 12.2% of the observations had an observed fouling rate of zero. Zero values can arise when the non-negative clipping of the net fouling rate is activated, including cases where the removal term offsets deposition; the observation-noise step can also produce values that are clipped to zero.

The proportional noise structure also means that larger fouling rates can have larger absolute errors. This was considered when interpreting the residual analysis.

3.7 Machine-Learning Input Variables

The primary machine-learning configuration used four predictors:

$$X = \left(\Re, T_f, \tau_w, F_p\right) \text{ specifically:}$$

1. Reynolds number
2. Film temperature
3. Wall shear stress
4. Fouling propensity index

The target variable was:

$$y = R_{f,obs}$$

expressed as fouling rate in m²K/kWh.

A secondary feature configuration was also investigated. In this case, the derived hydrodynamic variables were replaced by their primitive physical variables:

$X_{\text{primitive}} = \{v, D, \rho, T_f, F_p\}$ This comparison was designed to determine whether explicitly providing Reynolds number and wall shear stress offered an advantage over allowing the machine-learning algorithms to infer the relevant relationships from the underlying physical variables.

3.8 Data Preprocessing

The dataset was divided into training and testing subsets using an **80:20 random split**, with a fixed random state of 42.

For models requiring feature scaling, standardization was performed using:

$$z = \frac{x - \mu_x}{\sigma_x}$$

where x is an input variable, μ_x is its training-set mean and σ_x is its training-set standard deviation.

Linear Regression, MLP and SVR used standardized input variables. Random Forest did not require feature scaling because tree-based models are insensitive to the numerical scale of their predictors.

The target variable was also standardized for the MLP and SVR models because the fouling-rate distribution was right-skewed. Predictions were transformed back into the original fouling-rate units before evaluation.

For cross-validation, preprocessing was performed within the individual training folds using pipelines to prevent information from the validation folds from influencing the scaling parameters.

3.9 Machine-Learning Algorithms

Four regression algorithms were selected to provide comparison between linear and nonlinear modelling approaches.

3.9.1 Linear Regression

Linear Regression was used as the baseline model. The model assumes that the target can be represented as a linear combination of the input variables:

$y = \beta_0 + \sum_{j=1}^{p} \beta_j x_j$Its purpose was to establish the performance that could be obtained without explicitly modelling nonlinear relationships.

3.9.2 Random Forest Regression

Random Forest Regression was selected because it can represent nonlinear relationships and interactions between predictors without requiring an explicit mathematical form.

The final model contained **300 decision trees**. Other parameters retained the standard scikit-learn defaults, including unlimited tree depth, a minimum of two observations for splitting and a minimum leaf size of one.

Random Forest was also used for feature-importance analysis.

3.9.3 Multilayer Perceptron

The MLP model was implemented as a feed-forward artificial neural network containing two hidden layers:

$$64 \rightarrow 32$$

neurons.

The ReLU activation function and Adam optimization algorithm were used with the standard regularization parameter. The maximum number of iterations was set to 5,000.

The MLP was selected because neural networks can approximate complex nonlinear relationships and interactions between operating variables.

3.9.4 Support Vector Regression

SVR with a radial basis function (RBF) kernel was used as the principal kernel-based nonlinear model.

The principal settings were:

$$C = 10$$

and

$$\epsilon = 0.01$$

with the kernel width parameter γ set to the scikit-learn scale default.

SVR was selected because the RBF kernel can model nonlinear relationships while maintaining a controlled margin around the regression function.

3.10 Hyperparameter Selection

The reported model configurations were **not obtained through formal grid-search or Bayesian hyperparameter optimization**.

Instead, reasonable baseline values were selected based on standard configurations and computational stability. This distinction is important because the reported performance should not be described as the result of systematic hyperparameter optimization.

The main settings were:

Model	Principal settings
Linear Regression	Default configuration
Random Forest	300 trees
SVR	RBF kernel, C = 10, ε = 0.01, γ = scale
MLP	Two hidden layers: 64 and 32 neurons; ReLU; Adam; 5,000 maximum iterations

This approach was considered appropriate for the short-duration proof-of-concept study, although formal hyperparameter optimization would be appropriate in future work.

3.11 Model Evaluation

Three performance metrics were used.

3.11.1 Coefficient of Determination

The coefficient of determination was calculated as:

$$R^2 = 1 - \frac{\sum_{i=1}^{n} \left(y_i - \acute{y}_i \right)^2}{\sum_{i=1}^{n} \left(y_i - \overline{y} \right)^2}$$ where y_i is the observed value, $\hat{y}_i$ is the predicted value and $\underline{y}$ is the mean observed value.

Higher R^2 indicates stronger agreement between predictions and observations.

3.11.2 Root Mean Squared Error

$$\mathrm{RMSE}=\sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i-\acute{y}_i\right)^2}$$ RMSE gives greater weight to larger prediction errors.

3.11.3 Mean Absolute Error

$$\mathrm{MAE}=\frac{1}{n}\sum_{i=1}^{n}¿\, y_i-\acute{y}_i\vee ¿$$ MAE represents the average absolute difference between predicted and observed fouling rates.

3.12 Five-Fold Cross-Validation

Five-fold cross-validation was conducted to determine whether the model performance depended strongly on a particular train-test split.

The dataset was divided into five approximately equal subsets. In each iteration, four subsets were used for model training while the remaining subset was used for validation. The process was repeated until every subset had served as the validation set.

The mean and standard deviation of R^2, RMSE and MAE across the five folds were calculated.

This provided a measure of model performance stability in addition to the single 80:20 test-set evaluation.

3.13 Residual Analysis

Residuals were calculated as:

$$e_i=y_i-\hat{y}_i$$

where $e_i$represents the difference between the observed and predicted fouling rates.

Two graphical analyses were performed:

1. residuals versus predicted fouling rate;
2. residual-frequency distribution.

These analyses were used to identify systematic prediction bias, changes in error magnitude across the prediction range and unusually large prediction errors.

3.14 Feature-Importance Analysis

Two approaches were used to examine the contribution of the input variables.

First, the built-in impurity-based feature importance of the Random Forest model was calculated.

Second, **permutation importance** was calculated on the independent test set using 20 random permutations for each feature.

Permutation importance measures the reduction in predictive performance when the values of an individual feature are randomly shuffled while the remaining features are retained.

This provided a complementary assessment because conventional tree-based feature importance can be affected by relationships between predictors.

3.15 Physics-Consistency Analysis

A physics-consistency analysis was performed to determine whether the trained SVR reproduced the expected qualitative relationships embedded in the physics-informed formulation.

The expected relationships were:

Variable	Expected effect on fouling
Film temperature	Increase
Fouling propensity	Increase
Reynolds number	Decrease
Wall shear stress	Decrease

Rather than evaluating these relationships at only one arbitrary operating condition, 500 randomly selected background conditions were used. For each condition, 15-point sweeps were performed and the predicted response was evaluated across the sweep.

For each background condition, one variable was swept across its defined operating range while the remaining variables were held at their sampled values. Two measures were calculated:

1. **Net-direction consistency**, measuring whether the predicted value at the end of the sweep moved in the expected direction relative to the beginning.
2. **Step-wise consistency**, measuring the proportion of local changes that followed the expected direction.

This procedure provided a more robust assessment of whether the machine-learning model reproduced the qualitative trends imposed by the synthetic physics-informed formulation across different operating conditions. The analysis is therefore a consistency check against the adopted synthetic formulation, not independent validation against plant or experimental measurements.

3.16 Generalization to Unseen Operating Conditions

A separate extrapolation experiment was conducted to assess model performance outside the temperature range represented during training.

The 75th percentile of film temperature was used as the separation point. The models were trained using observations below approximately **566 K**, resulting in 2,250 training observations.

The highest-temperature quartile, consisting of **750 observations between approximately 566 and 620 K**, was withheld completely from model training and used as the generalization test set.

The resulting performance was compared with the random-split test performance.

This experiment was designed to distinguish between:

$$interpolation\,within\,the\,training\,distribution$$

and

$$extrapolation ¿ unseen\,operating\,conditions\,.$$

The distinction is particularly important for potential industrial deployment because a model can achieve strong performance when new observations resemble its training data while performing poorly when operating conditions move beyond the range represented in the training dataset.

3.17 Derived- versus Primitive-Variable Comparison

To assess the value of physics-based feature engineering, two input configurations were compared using the same train-test observations.

Configuration A: Derived variables

$$\left\{\Re\,, T_f, \tau_w, F_p\right\}$$

Configuration B: Primitive variables

$$\left\{\nu, D, \rho, T_f, F_p\right\}$$

The same four machine-learning algorithms were trained under both configurations.

The comparison was designed to determine whether explicitly calculating hydrodynamic quantities before machine learning improved predictive performance or whether the algorithms could recover the relevant relationships directly from the underlying physical variables.

3.18 Reproducibility and Computational Implementation

The modelling procedure was implemented in Python using NumPy and pandas for numerical and data-processing operations and scikit-learn for machine-learning algorithms, preprocessing, cross-validation and evaluation. The synthetic dataset contained 3,000 observations generated from the primitive-variable ranges defined in Section 3.2.

A fixed random seed of 42 was used for synthetic data generation, train-test splitting and stochastic model components where applicable. This ensured that the reported experiment could be reproduced using the same computational settings.

The study therefore followed the sequence:

$$Primitive\,variables \rightarrow viscosity \rightarrow \Re, \tau_w \rightarrow physics-informed\,fouling\,rate \rightarrow synthetic\,observatio$$

This methodology provides the basis for the model-performance results presented in Chapter Four.

4. RESULTS

4.1 Overview

This chapter presents the results obtained from the four machine-learning models developed for predicting crude-oil preheat-train fouling rate. The models evaluated were Linear Regression, Random Forest Regression, Multilayer Perceptron (MLP) and Support Vector Regression (SVR).

Model performance was assessed using an 80:20 random train-test split and five-fold cross-validation. Further analyses examined residual behaviour, feature importance, physics consistency, the effect of derived versus primitive variables, and model performance when predicting an unseen high-temperature regime.

4.2 Characteristics of the Synthetic Dataset

The final dataset contained **3,000 observations** generated from the physics-informed formulation described in Chapter Three.

The principal variables had the following distributions:

Variable	Minimum	Maximum	Mean	Standard deviation
Film temperature (K)	400.0	619.9	510.0	64.2
Reynolds number	2,319.6	117,815.2	30,326.2	22,333.5
Wall shear stress (Pa)	0.98	20.79	6.98	4.24
Fouling propensity index	0.50	1.50	0.999	0.290
Fouling rate (m^2K/kWh)	0.000	0.184	0.0173	0.0236

A proportion of the observations produced zero net fouling after the calculated net fouling rate was truncated at zero. This occurred when the removal contribution was sufficient to offset the deposition contribution. The dataset contained 365 zero-fouling observations, representing approximately 12.2% of the dataset.

The distribution demonstrates that the synthetic dataset covers a relatively broad range of thermal and hydrodynamic conditions rather than representing a single operating point.

4.3 Model Performance on the Random Test Set

The models were first evaluated using the independent 20% test subset. The results are presented in Table 4.2.

Model	R^2	RMSE	MAE
Linear Regression	0.7227	0.012823	0.009320
Random Forest	0.9778	0.003632	0.001904
MLP (ANN)	0.9883	0.002636	0.001491
SVR	0.9893	0.002522	0.001223

The nonlinear machine-learning models substantially outperformed the Linear Regression baseline.

Linear Regression achieved an R^2 of 0.7227, indicating that a simple linear representation captured a substantial portion of the variation but was unable to reproduce the nonlinear fouling relationship as accurately as the other models.

Random Forest improved the test-set R^2 to 0.9778. The MLP and SVR produced the highest performance, with R^2 values of 0.9883 and 0.9893, respectively.

SVR achieved the best overall random-test performance, with the lowest RMSE and MAE as well as the highest R^2.

4.4 Predicted versus Actual Fouling Rate

Figure 4.1 presents the predicted fouling rates against the actual values for the best-performing SVR model.

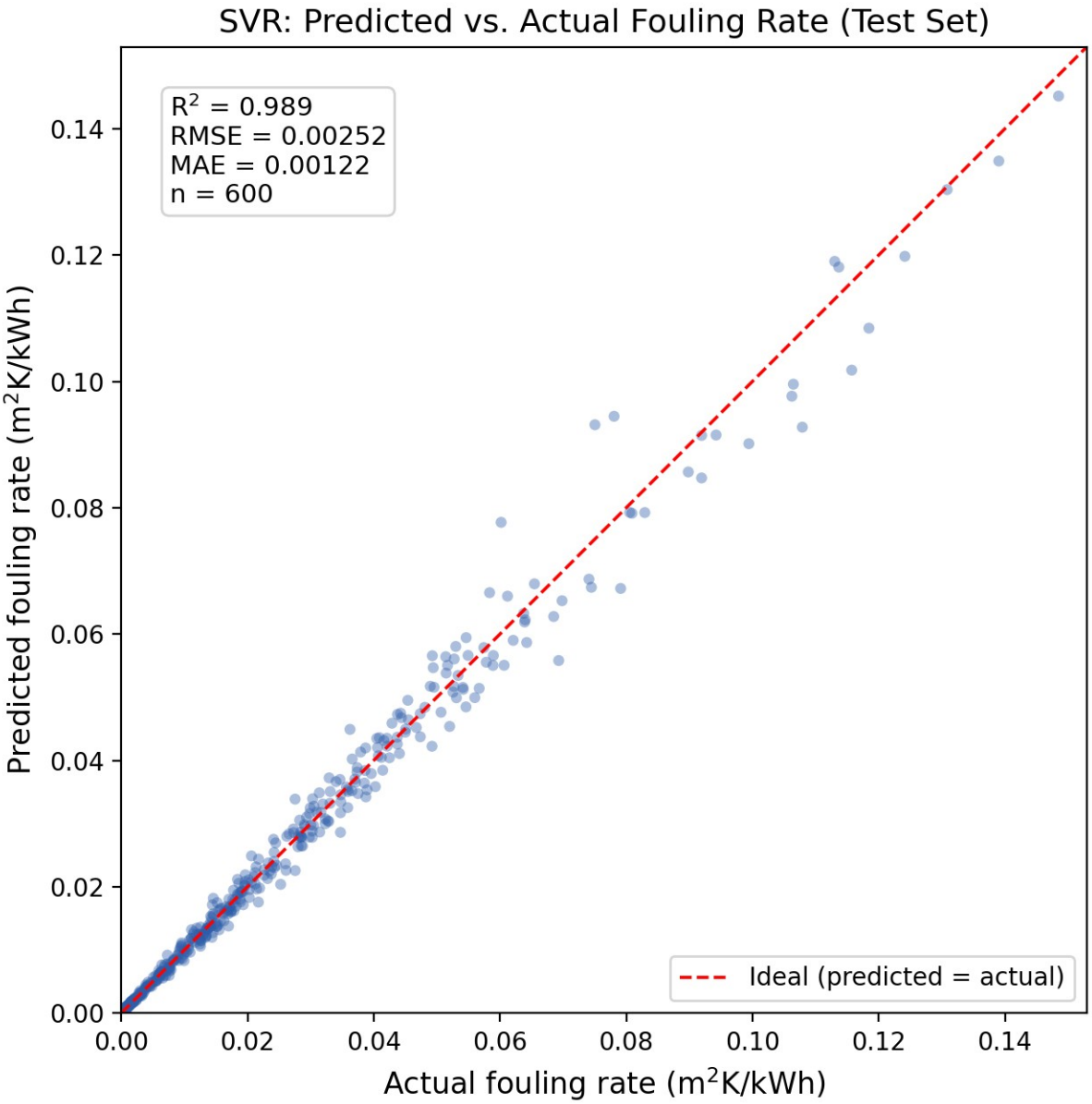


Figure 4.1

The majority of observations lie close to the ideal $y = x$ line, indicating strong agreement between predicted and actual fouling rates.

The concentration of points around the diagonal is consistent with the high test-set R^2 of 0.9893. Some deviations become more apparent at higher fouling rates, indicating that the model does not reproduce every high-fouling observation exactly.

The result demonstrates strong **interpolation performance within the distribution represented by the randomly divided dataset**. It should not, however, be interpreted as evidence that the model will maintain the same accuracy outside the training distribution. This issue is examined later through the dedicated generalization experiment.

4.5 Cross-Validation Results

Five-fold cross-validation was conducted to determine whether the observed model performance depended strongly on the particular random train-test split.

Model	Mean R^2	SD R^2	**Mean RMSE**	**SD RMSE**	**Mean MAE**	**SD MAE**
Linear Regression	0.7239	0.0037	0.012365	0.000357	0.009345	0.000163
Random Forest	0.9748	0.0035	0.003723	0.000191	0.001870	0.000044
MLP (ANN)	0.9860	0.0025	0.002767	0.000197	0.001480	0.000091
SVR	0.9870	0.0020	**0.002674**	**0.000183**	**0.001245**	**0.000035**

The cross-validation results closely followed the random-test results.

SVR again produced the highest mean R^2, at 0.9870 ± 0.0020, while MLP achieved 0.9860 ± 0.0025.

The relatively small standard deviations indicate that model performance was stable across the five folds. The agreement between the independent test-set results and cross-validation results provides evidence that the high interpolation performance was not caused solely by an unusually favourable random split.

4.6 Comparison of Model Performance

Figure 4.2 compares the random-test R^2values with the five-fold cross-validation results.

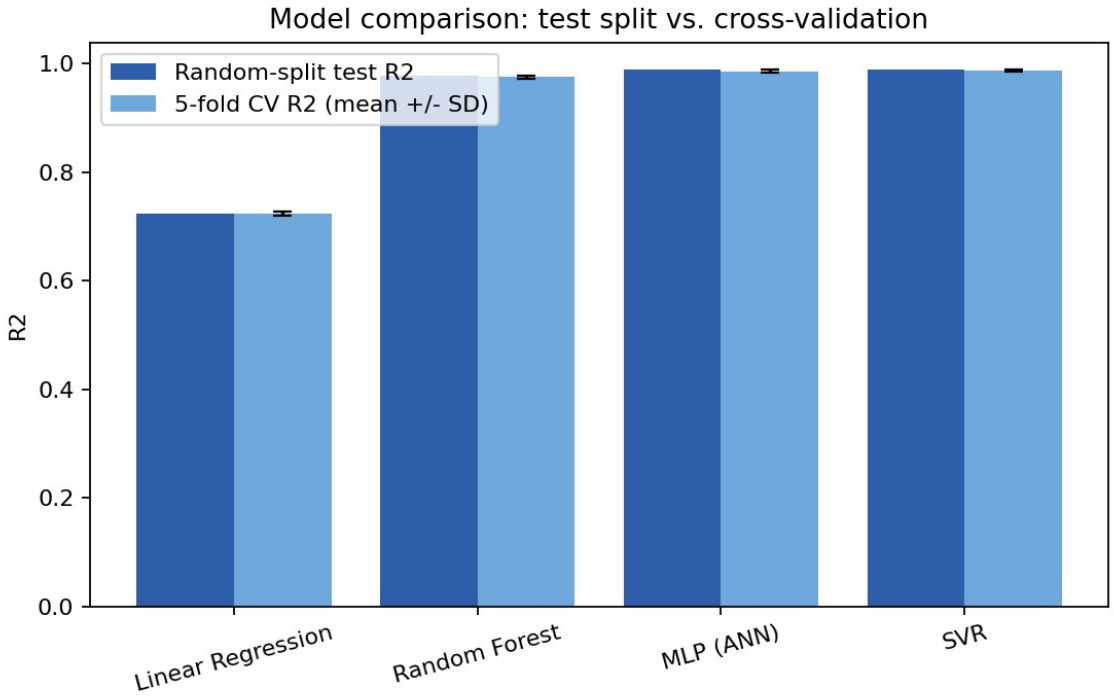


Figure 4.2

The two nonlinear models, SVR and MLP, consistently produced the strongest predictive performance. Random Forest also performed strongly but remained below the two kernel/neural-network approaches.

The ranking was broadly consistent:

$$SVR > MLP > Random\ Forest > Linear\ Regression$$

for both the random test and cross-validation results.

The close performance of SVR and MLP suggests that the underlying synthetic fouling relationship can be effectively represented using different nonlinear learning approaches.

4.7 Residual Analysis

Residual analysis was conducted using the SVR model, which produced the highest random-test R^2.

Residuals were defined as e = y_actual − y_predicted.

Figure 4.3 shows the residual-versus-predicted plot (left) and the residual distribution (right) for the SVR model.

$$e = y - \hat{y}$$

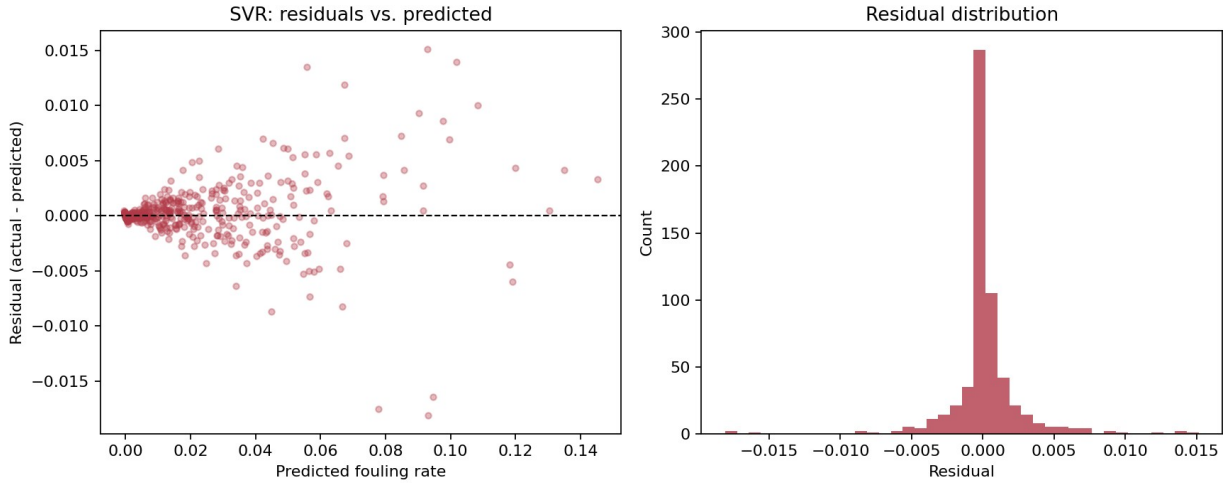


Figure 4.3

The residual-versus-predicted plot shows that most residuals are concentrated around zero. This indicates that the model does not exhibit a large systematic bias across the majority of the prediction range.

The residual distribution is strongly concentrated around zero, although a relatively small number of larger positive and negative residuals are present.

The increasing spread of residuals at higher predicted fouling rates suggests some degree of non-constant error variance. This is reasonable given that the synthetic noise was defined partly as a function of the magnitude of the fouling rate.

Therefore, although the model achieves a high R^2, the residual analysis indicates that prediction errors are not completely uniform across the full response range.

4.8 Random Forest Feature Importance

The Random Forest model was used to examine the relative contribution of the four predictors.

The permutation importance results were:

Feature	**Permutation importance**
Film temperature	**1.6683 ± 0.0865**

Feature	Permutation importance
Fouling propensity index	0.2522 ± 0.0351
Wall shear stress	0.2044 ± 0.0129
Reynolds number	0.0509 ± 0.0062

Film temperature was by far the most influential predictor.

Its permutation importance was substantially larger than those of the other variables, indicating that disrupting temperature information produced the largest deterioration in Random Forest predictive performance.

The same general ranking was obtained using the Random Forest's built-in feature importance measure.

This result is consistent with the strong temperature dependence introduced by the Arrhenius term in the physics-informed fouling formulation.

4.9 Derived versus Primitive Features

The effect of physics-based feature engineering was investigated by comparing two input configurations.

Configuration A used the derived hydrodynamic variables:

$$\left\{\Re, T_f, \tau_w, F_p\right\}$$

while **Configuration B** used the underlying primitive variables:

$$\left\{v, D, \rho, T_f, F_p\right\}.$$

The results were:

Model	Derived features	Primitive features
Linear Regression	0.7227	0.7183
Random Forest	0.9778	0.9773
MLP (ANN)	0.9883	0.9854
SVR	0.9893	0.9876

Model	Derived features	Primitive features

The derived-feature configuration produced a small improvement for every model.

The largest difference occurred for the MLP:

$$0.9883 - 0.9854 = 0.0029$$

while the SVR improvement was:

$$0.9893 - 0.9876 = 0.0017 .$$

The relatively small differences indicate that the nonlinear models were capable of recovering much of the relevant physical relationship from the primitive variables themselves.

Nevertheless, explicitly providing Reynolds number and wall shear stress offered a modest and consistent predictive advantage.

4.10 Physics-Consistency Analysis

The physics-consistency analysis evaluated whether the SVR reproduced the expected qualitative relationships across 500 randomly selected background conditions.

The results are shown in Table 4.6.

Variable	Expected direction	Net consistency	Step-wise consistency
Film temperature	Increase	**99.4%**	75.5%
Fouling propensity	Increase	**88.8%**	**88.8%**
Reynolds number	Decrease	**57.2%**	**57.4%**
Wall shear stress	Decrease	**97.4%**	**84.3%**

Film temperature showed the strongest net physical consistency, with 99.4% of background conditions producing the expected increase in predicted fouling rate.

Fouling propensity also showed predominantly expected behaviour, with 88.8% net consistency.

Wall shear stress exhibited 97.4% net consistency, supporting the expected relationship between increased shear and reduced fouling.

Reynolds number produced a substantially weaker result. Only 57.2% of background conditions exhibited the expected net decrease in predicted fouling rate, while 57.4% of local steps followed the expected direction.

Thus, the SVR should **not be described as universally physics-consistent**. Its learned response to Reynolds number was considerably less reliable than its responses to temperature, fouling propensity and wall shear stress.

This finding is particularly important because high overall predictive accuracy does not necessarily mean that the model has learned every underlying physical relationship correctly.

4.11 Generalization to an Unseen Hot Regime

The random train-test experiment evaluates interpolation because observations from the entire temperature range can appear in both training and test sets. A separate experiment was therefore conducted in which the highest-temperature quartile was completely withheld from training.

The models were trained on temperatures below approximately **566 K** using 2,250 observations and evaluated on 750 observations between approximately **566 and 620 K**.

Model	Random-test R^2	Unseen-hot-regime R^2
Linear Regression	0.7227	**-0.5159**
Random Forest	0.9778	**-0.0280**
MLP (ANN)	0.9883	**0.7495**
SVR	0.9893	0.5519

A substantial reduction in predictive performance occurred for all models.

Linear Regression and Random Forest produced negative R^2values on the unseen regime. A negative R^2indicates that the model performed worse than a baseline that simply predicts the mean of the test observations.

The nonlinear models performed better. MLP achieved an $R^2$of 0.7495, while SVR achieved 0.5519.

Interestingly, this reverses the ranking observed under random splitting:

$$SVR > MLP$$

during interpolation, but:

$$MLP > SVR$$

when extrapolating into the unseen high-temperature regime.

This demonstrates a clear distinction between **interpolation accuracy and extrapolation capability**.

4.12 Summary of Major Findings

The principal results of the study can be summarized as follows:

1. SVR produced the highest random-test performance, achieving an R^2 of 0.9893, RMSE = 0.002522 and MAE = 0.001223.
2. MLP produced very similar performance, with an R^2 of 0.9883.
3. Five-fold cross-validation confirmed the stability of the nonlinear models, with SVR achieving an R^2 of 0.9870 ± 0.0020.
4. **Film temperature was the dominant predictor** according to Random Forest permutation importance.
5. Derived hydrodynamic variables produced a **small but consistent improvement** over the primitive-variable configuration.
6. The SVR reproduced the expected effects of temperature, fouling propensity and wall shear stress in most background conditions.
7. **Reynolds-number behaviour was substantially less physically consistent**, with only approximately 57% directional consistency.
8. The excellent random-split performance did not translate directly to unseen high-temperature conditions.
9. **MLP generalized better than SVR to the unseen hot regime**, despite SVR being the best model under random splitting.

The results therefore indicate that machine-learning models can accurately reproduce the physics-informed synthetic fouling relationship within the sampled operating domain, but their performance can deteriorate substantially when they are required to extrapolate beyond the conditions represented during training.

5. DISCUSSION

5.1 Overview

The results presented in Chapter Four demonstrate that nonlinear machine-learning models can reproduce the relationships contained within the physics-informed synthetic fouling dataset with substantially greater accuracy than a linear baseline. However, the results also reveal important differences between **interpolation performance**, **physical consistency**, and **generalization to unseen operating conditions**.

The discussion therefore focuses not only on which model achieved the highest R^2, but also on why the models behaved differently, what the feature-importance results indicate, and what the findings imply for the development of machine-learning approaches for crude-oil preheat-train fouling prediction.

5.2 Superiority of Nonlinear Models

The most prominent finding was the performance difference between Linear Regression and the three nonlinear models.

Linear Regression achieved a test $R^2$of 0.7227, compared with 0.9778 for Random Forest, 0.9883 for MLP and 0.9893 for SVR.

This difference is consistent with the mathematical structure of the physics-informed fouling formulation. The target variable is influenced by an exponential temperature term and a nonlinear Reynolds-number term:

$$R_f \propto R\, e^{\beta} \exp\left(\frac{-E_a}{R\,T_f}\right)$$

Consequently, the relationship between the predictors and fouling rate is not adequately represented by a simple linear function.

SVR produced the highest random-test performance, achieving an R^2 of 0.9893, RMSE = 0.002522 and MAE = 0.001223.

This supports the use of nonlinear machine-learning techniques for the type of fouling relationship investigated in this study.

However, the comparison should be interpreted within the limits of the synthetic dataset. Since the target was generated from a known mathematical relationship, the high performance of the nonlinear models indicates their ability to approximate that relationship rather than proving superior performance on real refinery measurements.

5.3 Why SVR Performed Best Within the Sampled Domain

SVR achieved the highest test-set performance:

$$R^2 = 0.9893$$

with:

$$RMSE = 0.002522$$

and:

$$MAE = 0.001223.$$

Its five-fold cross-validation performance was similarly strong:

$$R^2 = 0.9870 \pm 0.0020.$$

The small difference between the random-test and cross-validation results suggests that SVR provided stable interpolation across the sampled operating conditions.

The likely explanation is that the RBF kernel allows SVR to represent nonlinear relationships without requiring an explicit mathematical form for the relationship between the inputs and fouling rate. This is particularly appropriate for the nonlinear temperature dependence and interactions present in the synthetic formulation.

The MLP performed almost identically on the random test set, with $R^2 = 0.9883$. The difference between the two models is therefore small and should not be interpreted as evidence that SVR is universally superior to neural networks.

Instead, the results indicate that **SVR was the strongest model under the specific data distribution and parameter configuration used in this study**.

5.4 Importance of Film Temperature

Film temperature was clearly the most influential predictor in the Random Forest analysis.

Its permutation importance was:

$$1.6683 \pm 0.0865$$

which was substantially greater than the values obtained for fouling propensity, wall shear stress and Reynolds number.

This result is consistent with the structure of the physics-informed fouling formulation. Temperature occurs within the Arrhenius term:

$$\exp\left(\frac{-E_a}{R T_f}\right).$$

Because this relationship is exponential, changes in temperature can produce comparatively large changes in the deposition component of fouling.

The result is also consistent with the physics-consistency analysis, where 99.4% of the sampled background conditions produced the expected net increase in fouling when temperature was increased.

Therefore, the machine-learning results and the underlying physical formulation provide mutually consistent evidence that temperature is the dominant factor controlling the simulated fouling response.

It is important, however, not to interpret the permutation importance value as a direct percentage contribution to fouling. Permutation importance measures the effect of disrupting a feature on model performance and therefore represents **predictive importance**, not a causal decomposition of the fouling mechanism.

5.5 Role of Fouling Propensity

The fouling propensity index was the second most important feature according to permutation importance:

$$0.2522 \pm 0.0351.$$

This is physically reasonable because the index directly scales the deposition term:

$$R_{deposit} \propto F_p.$$

The physics-consistency analysis also showed that 88.8% of the background conditions exhibited the expected increase in fouling as fouling propensity increased.

The less-than-perfect consistency indicates that the trained SVR occasionally produced local responses that did not follow the underlying relationship exactly. This is expected from a machine-learning approximation and demonstrates that overall prediction accuracy does not guarantee perfect recovery of every individual physical relationship.

5.6 Role of Wall Shear Stress

Wall shear stress had a permutation importance of:

$$0.2044 \pm 0.0129 .$$

Its influence is explained by its direct role in the removal term:

$$R_{removal} = \gamma \tau_w .$$

An increase in wall shear stress therefore reduces the predicted net fouling rate in the underlying formulation.

The physics-consistency analysis provided strong support for this relationship. Approximately 97.4% of the background conditions exhibited the expected net decrease in fouling with increasing wall shear stress.

This suggests that the machine-learning model captured the removal mechanism relatively well, although the 84.3% step-wise consistency indicates that the learned relationship was not perfectly monotonic at every local point.

5.7 The Reynolds-Number Finding

The Reynolds-number result requires particular attention.

Reynolds number had the lowest permutation importance:

$$0.0509 \pm 0.0062 .$$

More importantly, its physics-consistency result was substantially weaker than those of the other predictors. Only 57.2% of background conditions exhibited the expected net decrease in fouling, while 57.4% of individual local steps followed the expected direction.

This means that the SVR did **not** learn a consistently monotonic Reynolds-number response.

This result should not simply be classified as a modelling failure. Reynolds number was calculated from:

$$\Re = \frac{\rho v D}{\mu}$$

and therefore already contains information associated with velocity, density, tube diameter and temperature-dependent viscosity. Wall shear stress was also calculated from related flow variables.

Consequently, Reynolds number does not provide completely independent information to the model.

The low permutation importance suggests that, once the other predictors are available, disrupting Reynolds-number information has relatively little additional effect on the Random Forest's predictive performance.

The physics-consistency result reinforces this interpretation: the model was capable of achieving very high overall accuracy without learning a clean isolated Reynolds-number response.

This finding is valuable because it demonstrates that **global predictive accuracy can conceal weaknesses in individual feature-response relationships**.

5.8 Effect of Physics-Based Feature Engineering

Replacing the derived hydrodynamic variables with the underlying primitive variables produced only small reductions in model performance.

For SVR:

$$R^2_{derived} = 0.9893$$

compared with:

$$R^2_{primitive} = 0.9876.$$

For MLP, the corresponding values were:

$$0.9883$$

and:

$$0.9854.$$

The results suggest that the nonlinear models were capable of learning much of the hydrodynamic information directly from velocity, diameter and density, while the derived variables provided a small additional predictive advantage.

Nevertheless, the derived-variable configuration consistently produced slightly better results. This suggests that incorporating physically meaningful transformations such as Reynolds

number and wall shear stress can provide useful information to machine-learning models without completely determining their predictive capability.

The relatively small performance difference also indicates that the machine-learning algorithms were not simply dependent on manually engineered variables.

5.9 Interpolation versus Extrapolation

One of the most important findings of the study was the difference between random-split performance and performance on the unseen high-temperature regime.

SVR achieved:

$$R^2 = 0.9893$$

under random splitting, but this declined to:

$$R^2 = 0.5519$$

when tested on the unseen high-temperature regime.

MLP showed a smaller generalization gap:

$$R^2 : 0.9883 \rightarrow 0.7495 .$$

Random Forest declined from 0.9778 to -0.0280, while Linear Regression declined from 0.7227 to -0.5159.

These results demonstrate that random train-test splitting can give an optimistic impression of model performance when observations from the same operating range are present in both training and test sets.

The MLP's stronger performance on the unseen high-temperature regime is particularly interesting. Although it was slightly inferior to SVR under interpolation, it retained considerably more predictive capability during extrapolation.

This suggests that **the best model under conventional random validation is not necessarily the best model for deployment in situations involving previously unseen operating conditions**.

For a practical refinery fouling-prediction system, this distinction is important. Operating conditions can change over time, and a model may encounter temperatures, crude properties or flow conditions outside the range represented in its training data.

5.10 Implications for Industrial Application

The results suggest that machine learning could provide a useful framework for fouling-rate prediction, particularly when combined with physically meaningful process variables.

A potential future implementation could use routinely monitored process variables such as temperature, flow rate, pressure and fluid properties to estimate fouling risk continuously.

However, the present study does **not** demonstrate that the developed models are ready for direct refinery deployment.

Three conditions would need to be addressed first.

First, real experimental data are required.

The present dataset is synthetic. Real crude-oil fouling is affected by factors that are not fully represented by the simplified formulation, including crude composition, deposit chemistry, surface characteristics and operating history.

Second, independent external validation is required.

A model should be evaluated using measurements from operating conditions and, ideally, equipment that were not involved in its development.

Third, extrapolation must be controlled.

The generalization experiment demonstrated that model performance can deteriorate considerably outside the training distribution. A practical system should therefore identify when operating conditions fall outside the range for which the model has been validated.

5.11 Significance of the Study

Despite the synthetic nature of the dataset, the study provides several useful methodological findings.

First, it demonstrates a reproducible framework for combining physical modelling with machine learning.

Second, it shows that nonlinear models can reproduce a physics-informed fouling relationship with very high accuracy within the sampled domain.

Third, the study demonstrates that feature importance and physics-consistency analyses can reveal information that would not be apparent from R^2alone.

Finally, the unseen-condition experiment demonstrates the importance of testing model generalization rather than relying exclusively on random train-test performance.

The study therefore contributes not only a comparison of four machine-learning algorithms, but also an evaluation framework that considers predictive accuracy, physical behaviour and generalization.

5.12 Limitations

Several limitations should be considered when interpreting the findings.

Synthetic data

The most important limitation is that the dataset was generated computationally rather than obtained from laboratory experiments or an operating refinery. The models therefore have not been experimentally validated.

Simplified fouling mechanism

The adopted fouling formulation captures temperature-dependent deposition and shear-related removal but does not represent the full chemical and physical complexity of crude-oil fouling.

Illustrative parameters

The values of α, E_a, $\gamma$and related parameters were literature-informed illustrative values rather than parameters fitted to a specific experimental dataset.

Limited hyperparameter optimization

The models used reasonable baseline hyperparameters rather than systematically optimized configurations. Formal hyperparameter optimization could potentially improve performance.

Extrapolation limitations

The generalization experiment demonstrated that performance can decline substantially outside the training distribution. This limits the extent to which the current models can be interpreted as robust predictors of previously unseen operating conditions.

5.13 Future Research

Future work should focus primarily on replacing the synthetic dataset with experimental or industrial measurements.

A real dataset should ideally contain:

- film or wall temperature;
- bulk fluid temperature;
- flow velocity or flow rate;
- pressure;

- Reynolds number;
- wall shear stress or sufficient variables to calculate it;
- crude-oil physical properties;
- crude composition or fouling propensity indicators;
- measured fouling resistance or fouling rate;
- operating time/history.

The next stage should also investigate systematic hyperparameter optimization, uncertainty quantification and validation using completely independent operating periods or refinery units.

Physics-informed machine-learning approaches could additionally be investigated in which the physical fouling constraints are incorporated directly into model training rather than evaluated only after training.

5.14 Chapter Summary

The study found that nonlinear machine-learning models substantially outperformed Linear Regression when reproducing the physics-informed synthetic fouling relationship. SVR produced the strongest interpolation performance, achieving a test R^2 of 0.9893 and five-fold cross-validation of 0.9870 ± 0.0020.

Film temperature was the dominant predictive feature, while wall shear stress and fouling propensity also demonstrated meaningful influence. Reynolds number showed relatively low predictive importance and weak monotonic consistency, illustrating that high overall model accuracy does not necessarily imply perfect recovery of every physical relationship.

Most importantly, the unseen high-temperature experiment revealed a substantial generalization gap. SVR's R^2 decreased from 0.9893 under random splitting to 0.5519 under extrapolation, while MLP retained stronger performance at 0.7495.

The findings therefore support the feasibility of machine learning as a framework for physics-informed fouling prediction while emphasizing that **real-data validation and testing under unseen operating conditions are essential before industrial application**.

6. CONCLUSION, RECOMMENDATIONS, LIMITATIONS AND FUTURE WORK

6.1 Conclusion

This study investigated the application of machine learning models for predicting crude-oil preheat-train fouling rate using a physics-informed synthetic dataset. The study was motivated by the importance of fouling prediction in maintaining heat-transfer performance and supporting more effective operation and maintenance of crude-oil preheat trains.

An Ebert-Panchal-style fouling-rate formulation was used as the physical basis for synthetic data generation. The formulation incorporated fouling propensity, film temperature, Reynolds number and wall shear stress to represent the competing effects of fouling deposition and removal. The synthetic dataset contained 3,000 observations and was generated from primitive operating variables including film temperature, velocity, tube diameter, crude density and fouling propensity. Reynolds number and wall shear stress were subsequently calculated from these variables rather than independently sampled.

Four machine learning approaches were evaluated: Linear Regression, Random Forest, Support Vector Regression (SVR), and a Multi-Layer Perceptron (MLP) neural network. Model performance was assessed using an 80/20 random train-test split and five-fold cross-validation.

The results showed substantial differences between the linear and nonlinear models. On the random 80/20 test set, Linear Regression achieved an R^2 of 0.7227, while Random Forest, SVR and MLP achieved R^2 values of 0.9778, 0.9893 and 0.9883, respectively. The cross-validation

results followed the same general pattern, with mean R^2 values of 0.7239, 0.9748, 0.9870 and 0.9860 for Linear Regression, Random Forest, SVR and MLP, respectively. These results indicate that the relationship represented by the synthetic fouling model is strongly nonlinear and is better captured by the nonlinear machine learning approaches evaluated.

Among the models evaluated on the random 80/20 test split, SVR produced the highest predictive performance, with an R^2 of 0.9893, RMSE of 0.002522 and MAE of 0.001223. However, the additional hot-regime holdout experiment demonstrated that high random-split performance did not necessarily translate into reliable extrapolation. When models were trained on film temperatures between 400 and 566 K and evaluated on an unseen range of 566 to 620 K, the MLP achieved an R^2 of 0.7495, compared with 0.5519 for SVR, -0.0280 for Random Forest and -0.5159 for Linear Regression. Therefore, the results demonstrate that model performance depends strongly on how generalization is evaluated.

The feature-importance analysis identified film temperature as the most influential input for the Random Forest model, followed by fouling propensity, wall shear stress and Reynolds number. The permutation importance values were 1.6683, 0.2522, 0.2044 and 0.0509, respectively. This indicates that temperature contributed substantially more unique predictive information than the other evaluated variables within the synthetic dataset.

The comparison between derived hydrodynamic features and primitive operating variables also showed that incorporating physically derived variables provided a small but consistent improvement. For example, the SVR R^2 increased from 0.9876 using primitive variables to 0.9893 using the derived feature set. Similar smaller improvements were observed for Random Forest and MLP. This suggests that physics-based feature engineering can provide useful information to machine learning models, although the models were still able to recover much of the underlying relationship from the primitive variables.

The physics-consistency analysis further demonstrated that the model did not reproduce every expected physical trend equally well. The predicted response generally followed the expected trends for film temperature, fouling propensity and wall shear stress. However, the Reynolds-number sweep produced a substantially weaker and more inconsistent response, with approximately 57% of tested cases following the expected direction. This result is consistent with the relatively low unique permutation importance of Reynolds number. It also demonstrates the importance of evaluating machine learning models against physical expectations rather than relying exclusively on statistical performance metrics.

Overall, the study demonstrates the methodological potential of combining a physics-based fouling formulation with machine learning for crude-oil preheat-train fouling prediction. However, because the investigation was based on synthetic rather than plant-measured data,

the results should be interpreted as a proof-of-method rather than evidence of immediate refinery deployment readiness.

6.2 Recommendations

Based on the findings of this study, the following recommendations are made:

1. **Machine learning should be investigated as a complementary approach to conventional fouling models.**
   The strong performance of the nonlinear models within the synthetic data distribution demonstrates that machine learning can represent complex nonlinear relationships between operating conditions and fouling rate.

2. **Model evaluation should extend beyond random train-test splitting.**
   Random splitting produced very high predictive performance for the nonlinear models, but the hot-regime holdout demonstrated substantially weaker extrapolation. Future studies should therefore include time-based, operating-regime-based or temperature-based holdout tests where appropriate.

3. **Physics-based features should be considered during model development.**
   Reynolds number and wall shear stress are physically meaningful quantities derived from operating conditions. Although their inclusion produced only modest improvements in this study, physics-based feature engineering can help connect machine learning predictions to established transport and fouling mechanisms.

4. **Film temperature should receive particular attention in future data collection and modelling.**
   Film temperature was the dominant feature in the permutation-importance analysis. Reliable temperature measurements and appropriate representation of thermal conditions are therefore important for developing practical fouling prediction models.

5. **Real refinery or pilot-plant data should be used for model validation before practical implementation.**
   Synthetic-data performance cannot establish how well a model will perform under actual crude-oil operating conditions. Models developed from synthetic data should therefore be treated as preliminary until validated against independent experimental or industrial datasets.

6. **Model development should incorporate uncertainty estimation.**
   A practical fouling prediction system should provide information about prediction uncertainty, particularly when operating conditions fall outside the range represented in the training data.

6.3 Limitations of the Study

Several limitations should be considered when interpreting the results.

First, the dataset used in this study was synthetically generated rather than obtained from an operating refinery or controlled experimental campaign. Consequently, the statistical relationships learned by the models reflect the assumptions embedded in the synthetic fouling formulation.

Second, the coefficients used in the Ebert-Panchal-style formulation were illustrative literature-informed parameters rather than parameters fitted to a specific crude-oil preheat-train dataset. The resulting values therefore should not be interpreted as plant-specific fouling kinetics.

Third, the primitive variables were sampled from simplified distributions. Real refinery operating variables are not generally independent and uniformly distributed. Temperature, velocity, density, viscosity, pressure, crude composition and heat-exchanger operating conditions can exhibit complex correlations and temporal variation.

Fourth, the synthetic model does not explicitly represent the full range of mechanisms that can contribute to crude-oil fouling. Actual fouling behaviour can depend on crude composition, deposit chemistry, surface condition, residence time, heat-transfer history, operating transients and other factors that were not explicitly included.

Fifth, the friction-factor relationship used during synthetic data generation was simplified. The generated Reynolds-number range extended to values around 2,300, meaning that not all observations fall comfortably within the turbulent regime for which the selected Blasius-type relationship is normally applied. This limits the physical generality of the synthetic dataset.

Sixth, the machine learning hyperparameters were selected manually for model stability and performance rather than through a systematic hyperparameter-optimization procedure. Consequently, the reported results should not be interpreted as an exhaustive search for the optimal configuration of each algorithm.

Finally, the study did not perform external validation using an independent real-world dataset. The high R^2 values obtained during random splitting therefore demonstrate performance within the synthetic data distribution rather than guaranteed performance under unseen industrial conditions.

6.4 Future Work

Future research should extend this study from synthetic-data modelling toward experimental and industrial validation.

The first priority should be the collection or acquisition of real crude-oil preheat-train operating data. Such a dataset should ideally contain temperature, flow rate, pressure, viscosity or viscosity-related information, crude properties, heat-transfer measurements, operating duration and measured fouling resistance or fouling rate.

Future studies should also investigate additional machine learning algorithms, including gradient-boosting methods and other ensemble approaches. These models could be compared with SVR and MLP using identical validation procedures.

A second important direction is **time-dependent fouling prediction**. The present study predicts fouling rate from operating variables, whereas real preheat-train fouling evolves over time. Sequential models could therefore be investigated for predicting future fouling resistance or the progression of fouling under changing operating conditions.

Future work should also incorporate **uncertainty quantification**. Prediction intervals or probabilistic models would provide additional information about the reliability of predictions and help identify operating conditions where the model is extrapolating beyond its training domain.

Another useful development would be the incorporation of additional physical variables and fouling mechanisms. These could include crude composition, surface temperature, bulk temperature, pressure, viscosity, heat flux, residence time, tube geometry and deposit characteristics.

Finally, future research should investigate **physics-informed or hybrid machine learning models** in which established fouling equations are combined directly with data-driven models. Such approaches may provide a better balance between predictive accuracy, physical consistency and interpretability than purely data-driven modelling.

6.5 Chapter Summary

This chapter presented the conclusions, recommendations, limitations and future directions of the study. The investigation demonstrated that nonlinear machine learning models substantially outperformed Linear Regression within the synthetic dataset, with SVR obtaining the highest R^2 on the random test split. However, the hot-regime holdout demonstrated that strong interpolation performance does not guarantee reliable extrapolation. The MLP showed the strongest performance among the evaluated models in this unseen high-temperature regime.

Film temperature was identified as the dominant predictive feature, while the comparison of derived and primitive variables showed a small advantage for physics-based feature

engineering. The physics-consistency analysis also highlighted the need to evaluate machine learning predictions against expected physical behaviour.

The central conclusion is therefore that **machine learning shows methodological potential for crude-oil preheat-train fouling-rate prediction, but validation using real operating data is necessary before the approach can be considered suitable for practical refinery application**.

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1